

# Brief Guide: Modbus 4-20mA Distance Sensor

## 1. Key Technical Specifications

- **Measurement Range:** 0.2...8 m (hardware blind zone is less than 200 mm).
- **Maximum Achievable Resolution:** 1 mm (provided by built-in digital processing).
- **Power Supply Voltage:** 8...28 V DC (VDC).
- **Power Consumption:** Not exceeding 3 W.
- **Communication Interface:** RS-485 (two-wire, half-duplex, default port settings: 9600 bps, 8N1).
- **Analog Output:** Active 4–20 mA current loop (internally powered, 12-bit DAC, load resistance not exceeding 250 ohms).

## 2. Electrical Connection (Pinout and Wire Colors)

Wiring must be performed strictly while the power supply is switched off. For long communication lines, it is recommended to install a 120-ohm termination resistor at the ends of the RS-485 line.

| Pin | Wire Color | Signal    | Purpose                              |
|-----|------------|-----------|--------------------------------------|
| 1   | Red        | 8...28VDC | Power supply positive (+)            |
| 2   | Blue       | 0V        | Power supply negative / Common (-)   |
| 3   | Black      | RS485-A   | RS-485 communication line (A / RxD+) |
| 4   | Yellow     | RS485-B   | RS-485 communication line (B / TxD-) |
| 5   | Green      | AO+       | Current loop output (+)              |
| 6   | White      | AO-       | Current loop output (-)              |

### 3. Modbus Register Map (Holding Registers)

Data exchange is performed using standard Modbus RTU functions: 03h (read), 06h, and 10h (write). Parameter changes take effect immediately after writing.

| Registers     | Parameter      | Access | Type           | Description / Default Values                                                                                                                |
|---------------|----------------|--------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>0, 1</b>   | Measurements   | R      | UInt16         | <b>Reg0:</b> Distance (mm). <b>Reg1:</b> Reflected signal level from the measurement module (arbitrary units).                              |
| <b>2, 3</b>   | Measurements   | R      | Int16 / UInt16 | <b>Reg2:</b> Internal temperature (°C, Signed Int16). <b>Reg3:</b> Calculated output current (Unsigned UInt16, ×100 mA).                    |
| <b>10, 11</b> | Comm / Polling | R/W    | UInt16         | <b>Reg10:</b> Modbus Address (1..247, default: 1). <b>Reg11:</b> Sensor poll rate (1..100 Hz, default: 10).                                 |
| <b>12, 13</b> | Filtering      | R/W    | UInt16         | <b>Reg12:</b> Median filter window size (1..15 measurements, default: 10). <b>Reg13:</b> EMA smoothing coefficient (0..100%, default: 20%). |
| <b>14, 15</b> | Calibration    | R/W    | UInt16         | Distance for reference points <b>D1</b> (Reg14, default: 500 mm) and <b>D2</b> (Reg15, default: 5000 mm).                                   |
| <b>16, 17</b> | Current Range  | R/W    | UInt16         | Distance boundaries <b>X1</b> for 4 mA (Reg16, default: 250) and <b>X2</b> for 20 mA (Reg17, default: 5000).                                |
| <b>18, 19</b> | Output DAC     | R/W    | UInt16         | Calibration DAC codes <b>Y1</b> for 4 mA (Reg18, default: 819) and <b>Y2</b> for 20 mA (Reg19, default: 4095).                              |
| <b>20</b>     | Operation Mode | R/W    | UInt16         | <b>0:</b> normal operation. <b>1, 2:</b> 4 / 20 mA current calibration. <b>3, 4:</b> reference point capture (D1 / D2).                     |

## 4. Two-Point Calibration of the Measurement Tract

Calibration compensates for systematic errors in the measurement tract. It is recommended to choose calibration points that are as far apart as possible within your physical measurement zone.

1. **Point 1:** Place a target in front of the sensor at a stable, known distance (e.g., 500 mm). Write this distance value: `Reg14 = 500`. Trigger the first capture by writing the command: `Reg20 = 3`.
2. **Point 2:** Move the target to a second known distance (e.g., 5000 mm). Write this distance value: `Reg15 = 5000`. Trigger the second capture by writing the command: `Reg20 = 4`. The device will calculate the coefficients and save them to the non-volatile memory.
3. **Completion:** Write the command `Reg20 = 0` to the control register to return the device to normal operation mode.

## 5. 4-20 mA Analog Output Configuration and Calibration

1. **Scaling:** Define the distance range corresponding to the minimum and maximum output current by configuring `Reg16` (X1 — 4 mA distance) and `Reg17` (X2 — 20 mA distance).
2. **Current Loop Calibration:** Connect a milliammeter in series with the analog output. Ensure that the load resistance in the current loop does not exceed 250 ohms.
  - *4 mA Calibration:* Write the command `Reg20 = 1`. Modify the value of register `Reg18` until the milliammeter reads exactly 4.00 mA.
  - *20 mA Calibration:* Write the command `Reg20 = 2`. Modify the value of register `Reg19` until the milliammeter reads exactly 20.00 mA.
3. **Completion:** Write the command `Reg20 = 0` to save all settings to the non-volatile memory.

## 6. Installation and Operation Requirements

- The sensor does not have to be installed strictly vertically. However, for maximum measurement accuracy, the laser beam must be perpendicular to the measured target surface.
- Ensure there are no foreign objects in the beam path (such as tank walls, internal ladders, pipes, or floating debris).
- Avoid condensation and dirt buildup on the optical window.
- When operating the device outdoors, install a protective cover to shield the optical window from weather precipitation and direct sunlight.